

Research Article

Vortex-Induced Vibration of Long Suspenders of a Long-Span Suspension Bridge and Its Effect on Local Deck Acceleration Based on Field Monitoring

Xun Su ¹, Jianxiao Mao ¹, Hao Wang ¹, Hui Gao ¹, Xiaoming Guo ¹,
and Hai Zong ^{2,3}

¹Key Laboratory of Concrete and Prestressed Concrete Structures of Ministry of Education, Southeast University, Nanjing 211189, China

²School of Transportation, Southeast University, Nanjing 211189, China

³Nanjing Highway Development (Group) Co., Ltd., Nanjing 210002, China

Correspondence should be addressed to Jianxiao Mao; jianxiao@seu.edu.cn

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As the main structural component, the possibility of wind-induced vibration, especially vortex-induced vibrations (VIVs), is greatly increased due to the shape and structural characteristics of the long suspenders. To investigate the full-scale wind-induced vibration of the long suspenders of a long-span suspension bridge with a main span of 1418 m, the long-term vibration-based monitoring system was established. Based on the recorded structural health monitoring (SHM) data, the corresponding wind conditions and the vibration characteristics of long suspenders with different diameters and tensions are investigated. Furthermore, modal parameters including frequencies and damping ratios of long suspenders are identified and tracked during the VIV period. The relationship between the shedding frequency of long suspenders and the corresponding wind speed is studied. Results show that the VIVs with frequencies ranging from 8 Hz to 20 Hz were observed continuously across a wide range of wind speeds in both sets of long suspenders. Due to the relatively low modal damping, significant vortex characteristics and lock-in phenomena can be expected on the long suspenders. A new frequency-adjustable Stockbridge damper is employed to suppress multimodal VIVs in the long suspenders. The effectiveness of Stockbridge damper is verified through field application and comparative analysis. Finally, the effect of long suspender VIVs on local deck vibration is discussed, and it is clarified that the bridge deck vibration is mainly caused by multimodal VIVs of the long suspenders, rather than by external loads such as vehicles and wind. The study endeavors to provide a case to progress the identification, assessment, and control of long suspender VIVs in similar long-span bridges.

1. Introduction

Suspension bridges have become one of the preferred bridge types for long-span bridges with the advantage of excellent spanning capacity and lightweight, aesthetically pleasing structures [1, 2]. As an integral element of suspension bridges, the long suspenders, characterized by a lower degree of rigidity and damping, exhibit a susceptibility to wind-induced vibrations. Excessive vibrations have been witnessed traversing the long suspenders of numerous distinguished suspension bridges, including the Great Belt East Bridge in Denmark [3],

Akashi Kaikyo Bridge in Japan [4], and the Xihoumen Bridge in China [5]. Excessive wind-induced vibration could raise concerns about the heavy fatigue of these long suspenders. Moreover, it creates a visual unpleasantness for the person driving the car. Consequently, the assessment of vibration suitability has assumed significant importance [6, 7].

It is imperative to conduct further analysis on the underlying causes of vibrations in long suspenders to implement efficient damping measures. Kashima et al. [3] conducted an examination on the abnormal vibrations of the long suspenders of the Akashi Kaikyo Bridge in Japan. The

vibrations were found to be the result of wake-induced flutter arising from aerodynamic disturbances present between the vertical suspenders. The high-amplitude vibrations occur more frequently at lower wind speed ranges (6 to 15 m/s) for the Great Belt East Bridge [3]. It is presumed that wind-ice or storm-induced galloping vibrations are responsible, although the precise mechanism remains unclear [8]. Chen et al. [9] showed through numerical simulations and wind tunnel tests that the long suspender vibration of the Xihoumen Bridge could be attributed to the wake of the towers. Cantero et al. [10] found that VIVs occurred in the long suspenders of the Hardanger Bridge in Norway through long-term monitoring. Evidently, the long suspender vibrations of long-span suspension bridges are diverse and the mechanism of generation is exceedingly intricate.

Vortex-induced vibrations (VIVs) are inherently nonlinear, self-regulating, multi-degree-of-freedom motions resulting from the fluid-structure interactions [11–13]. The vibrations usually excite the higher modes of the suspenders with relatively low amplitudes of displacement [14–16]. Although such vibrations may not cause devastating damage to the long suspenders, they can generate additional stresses and severely shorten the endurance of the suspenders [10, 17]. Since the 1980s, numerous wind tunnel tests and numerical simulations [18–22] have been conducted, aiming to examine the circumstances, vibrational characteristics, and excitation mechanisms of long suspender VIVs. They have significantly advanced the study of wind effects on cable structures [23–25]. However, variations in ambient and operating conditions such as traffic volumes and wind fields can affect the trend of VIV development [25, 26]. Such unforeseeable factors dictate the need for continuous monitoring of VIVs during the operation of long-span bridges to achieve accurate vibration suitability assessments.

Structural health monitoring (SHM) stands as the relatively direct means to comprehend the progression of engineering structures and guarantee their security [27–29]. In this regard, the SHM system has been adopted for most of the long-span bridges, providing convenience for capturing and analyzing the response characteristics of suspension VIVs under different operating conditions [30–32]. Additionally, with insights gained from monitoring, numerous aerodynamic and mechanical interventions have been employed to alleviate VIVs [33–37]. The successful experiences of vibration damping on long suspenders are difficult to simply replicate between different bridges [7, 25]. Therefore, further research based on monitoring is needed to accurately assess the hazardous vibration of long suspenders. Furthermore, the effect of long suspender vibration on the acceleration of the adjacent deck has rarely been acknowledged. By undertaking this endeavor, the extent to which deck acceleration is affected by long suspender vibrations can be further clarified.

This study evaluates the response of VIVs of different long suspenders of the Nanjing Qixiashan Yangtze River Bridge, namely, NQYRB, based on long-term monitoring data of the VBM. The relationship between the acceleration and the damping ratio of VIVs and the correlation between the frequency and the mean wind speed are analyzed. The effect of long suspender vibration on the local deck acceleration is also discussed. The study is organized as follows: in

Section 2, the NQYRB and corresponding VBM system are introduced. The high-frequency vibration response of long suspenders is discussed in Section 3. The damping measures for the long suspender VIVs are analyzed and evaluated, and the effect of long suspender VIV on local deck acceleration is discussed in Section 4. In Section 5, the conclusions drawn from this study are presented.

2. Engineering Background

2.1. Description of the NQYRB. The Nanjing Qixiashan Yangtze River Bridge (NQYRB) is a continuous three-span suspension bridge in China (Figure 1(a)). The lengths of the three spans are 576.2 m, 1418 m, and 481.8 m, respectively. The total number of suspension points for the NQYRB is 276, with each suspension point being equipped with two suspenders. The suspenders consisting of parallel steel wires are connected to the cable clamps and the steel box girders with pin hinges. They are divided into three categories: the first type is the displacement-limiting suspenders (DLS) at the top of the transition pier; the second type is special suspenders (SS) with higher forces and special requirements for deformation on the side of the tower and the adjacent displacement-limiting suspenders; the third type is ordinary suspenders (OS). Among them, there are two types of long suspenders: ordinary suspenders and special suspenders.

Since the bridge was opened in December 2012, the long suspenders have frequently exhibited wind-induced vibration at normal wind speeds. To mitigate the vibrations and enhance the restraint, three spacers were installed on each set of suspenders. However, excessive vibrations were still monitored in several sets of long suspenders near the pylons. A short-term field monitoring of the long suspenders revealed that the amplitude of the acceleration response reached 160 m/s^2 and the vibration frequency was between 10 Hz and 20 Hz [38]. Based on the wind tunnel test and the vibration characteristic analysis caused by the wake flow of the pylons and the wake flow between the cable strands, it is found that the vibration is low frequency and large amplitude [5, 9]. This further clarifies the VIV that occurs in these suspension bridge long suspenders.

2.2. Designed Monitoring Plan for Long Suspenders of the NQYRB. To further investigate the properties and the excitation mechanisms of the VIVs of the long suspenders, a VBM system has been established to directly monitor the acceleration of the long suspenders and the surrounding wind environment. Four accelerometers are affixed on the long suspenders, and other four accelerometers are mounted on the deck to monitor the in situ acceleration of the deck and suspenders at a sampling frequency of 200 Hz. For the accelerometer on the suspenders, the maximum measurement range is 500 m/s^2 . For the pickup on the main girder, the maximum measurement range is 20 m/s^2 . The ultrasonic anemometer is positioned in the vicinity of the monitoring long suspenders with a sampling frequency of 10 Hz. The monitoring data are stored at hourly intervals to save storage space. The configuration of the VBM is illustrated in Figure 1(b). The investigated long suspenders are referred to as S2 and S3, where S3 is a set of ordinary suspenders and S2

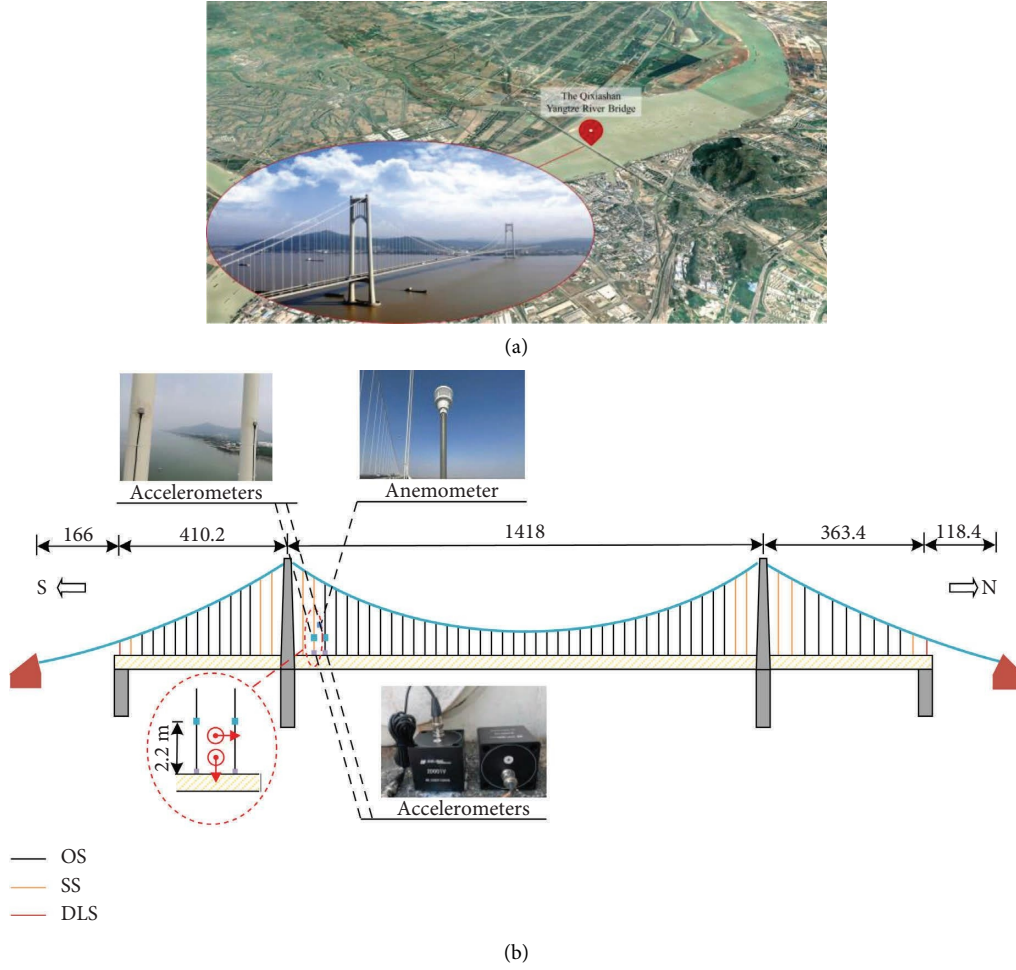


FIGURE 1: Demonstration of the NQYRB and VBM systems (unit: m). (a) The investigated long-span suspension bridge. (b) Introduction of the VBM system.

is special suspenders. The dimensions and characteristics are shown in Figure 2 and Table 1.

The monitored acceleration and wind data from September 08, 2022, to April 27, 2023, were utilized to investigate the characteristics and incentives of the VIVs of long suspenders. The continuous data set was split into a 10-minute segment. The vibrations of two long suspenders at the identical lifting point are markedly interrelated due to the synergistic effect of the spacers, and only one long suspender is selected for analysis. For the collected data, several faults were first excluded (1) anomalous values, i.e., 10-minute average wind speeds greater than 50 m/s or RMS of long suspender vibrations exceeded 500 m/s²; (2) sensor failures, including noise interference, measurement failures, and loss of accuracy; and (3) the peak that occurs when the ratio of the maximum deviation of the recorded signal to the standard deviation exceeds 30.

2.3. Preliminary Analysis of the Monitored Data

2.3.1. Wind Environment Analysis near the S2 and S3. The wind field distribution contributes significantly to the high-frequency vibration of the long suspenders. The

anemometer was mounted in the vicinity of the investigated long suspenders. A free-field wind of approximately constant magnitude can be expected in the vertical dimension. To eliminate the large variability induced by the gust wind, the 10-minute average was used to represent the instantaneous wind speeds. Figure 3 shows the wind rose diagram based on wind speed and direction data received from the ultrasonic anemometer. It is seen that the predominant wind direction in the vicinity of the long suspenders during this period was south-east, although not being perpendicular to the longitudinal axis of the bridge. In addition, wind speeds above 9 m/s always come from the south-west.

The turbulence characteristics were also investigated for each 10-minute wind section. As shown in Figure 4(a), the turbulence intensity decreases as a whole with the increase of the average wind speed and gradually tends to be stable. When the average wind speed is less than 3 m/s, the distribution of turbulence intensity is more discrete and random. The tendency of the turbulence integration scale with wind speed is shown in Figure 4(b). It can be found that as the mean wind speed increases from 0 m/s to 6 m/s, the turbulence integration scale gradually increases, and it is mainly distributed below 500 m, but there are also some large-scale vortices.

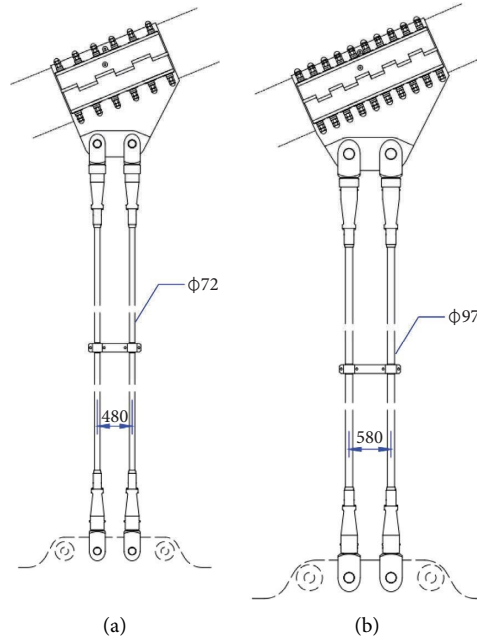


FIGURE 2: Ordinary and special suspenders and the detailed view (unit: mm). (a) S3. (b) S2.

TABLE 1: The dimensions and characteristics of S2 and S3.

Number	Diameter (m)	Tension (kN)	Equivalent volumetric weight (kN/m ³)	Length (m)	Distance (m)	Fundamental frequency (Hz)
S2	0.097	1087.25	82.42	145.17	0.58	0.610
S3	0.072	822.77	85.35	138.66	0.48	0.797

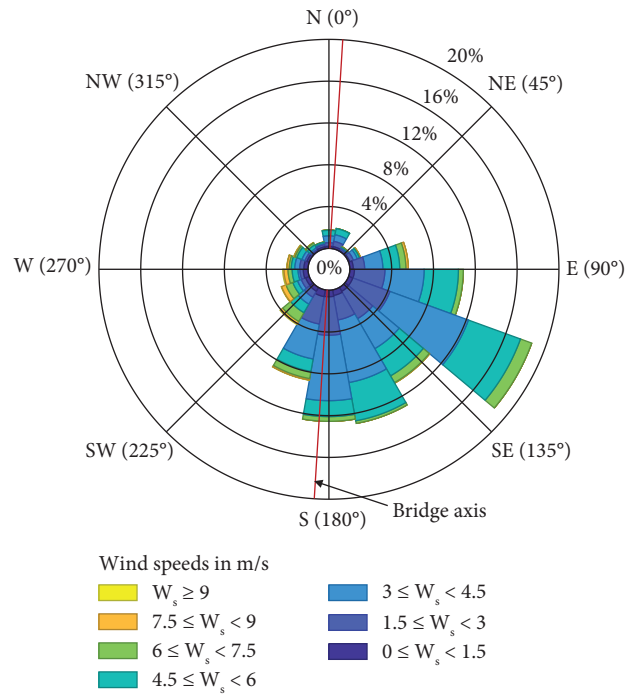


FIGURE 3: Rose diagram of the wind speed and direction.

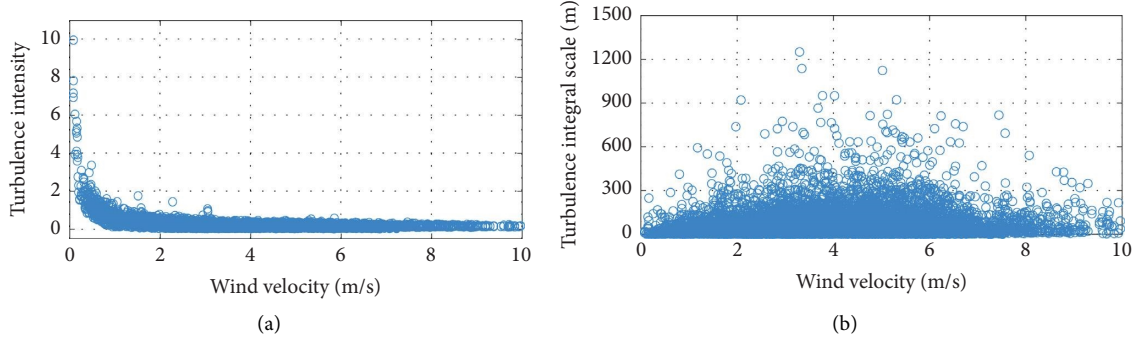


FIGURE 4: Analysis of turbulence characteristics. (a) Turbulence intensity. (b) Turbulence integral scale.

2.3.2. Wind-Induced Vibration Monitoring of Long Suspenders. Time-frequency features provide a clearer description of the nonstationary signal characteristics than time- and frequency-domain signals. The typical in-plane (i.e., parallel to the direction of the bridge axis) vibration acceleration time history for S3 and S2 and the corresponding time-frequency energy diagrams are presented in Figures 5 to 6. In the time-frequency energy diagram, high energy levels are shown in dark red, low-energy levels fade to yellow and green, and no energy levels are indicated as blue. As shown in Figures 5(a) and 6, the maximum vibration acceleration of the S3 is close to 180 m/s^2 , and the in-plane vibration acceleration of S2 is close to 20 m/s^2 in the VIV samples. In Figure 5(b), the VIV of S3 starts to develop at the second hour, when the wind speed is close to 6 m/s and the wind direction is about 120° . In Figure 5(b), the S3 generated VIVs from the 2 h when the wind speed is close to 6 m/s and the wind direction is around 120° . Subsequently, sustained disturbances from wind loads led to the full development of the VIVs. As a whole, VIVs occurred in the S3 when wind speeds ranged from 3.5 m/s to 6.5 m/s and wind directions ranged from 65° to 130° . In addition, a wide frequency band vibration from 5 Hz to 15 Hz emerged in S3, with several dense peaks near 10 Hz . It is clear that at any given time, the VIVs of the long suspenders are not centralized in a single mode and several modes would be stimulated at the same time.

The VIVs generated at wind speeds up to 4 m/s of S2 in Figure 6(b). It continued to develop and gradually climbed to a peak with wind speeds from 3 m/s to 6 m/s and wind directions from 110° to 160° . Different from S3, S2 has a narrower frequency band spread from 5 Hz to 12 Hz and a smaller number of peaks. This is primarily due to the fact that S2 has greater tension and stiffness. It also leads to a smaller acceleration of S2. The vibration of the long suspenders shows a nonstationary trend due to the time-varying character of incentives, resulting in a continuous change of the dominant modes. From the typical samples of the two long suspenders, no obvious regularity with the wind speed and direction could be detected. Therefore, the mechanism and characteristics of the wind-induced vibrations should be further investigated by long-term monitoring data of long suspenders.

3. High-Frequency Vibration Response of Long Suspenders

3.1. Frequency Characteristics of Long Suspender VIVs. Since the pulsating properties of the wind, the vortex shedding frequency also fluctuates constantly, resulting in nonsmooth excitation on the long suspenders. To identify the vibration response of the long suspenders during VIVs, a signal decomposition procedure with automatic peak pickup and empirical wavelet transform (EWT) is proposed (Figure 7). The target frequency and bandwidth first need to be determined in advance to effectively decompose the frequency of the long suspender. The fast Fourier transform (FFT) is used to determine the fundamental frequency of the suspender. The modal frequency of the VIVs of the long suspender is close to integer multiples of the fundamental frequency, from which the threshold of the frequency band is initially set. In the second step, the reference position of the target mode is determined by sliding the frequency band to select the peak point of the normalized PSD. It is then determined whether only one mode is included, if not, a local maximum needs to be selected in that frequency band. Compare it with an integer multiple of the fundamental frequency, then proceed to the next step. Finally, based on the fact that empirical wavelet transform (EWT) performs relatively well in dealing with nonlinear and nonsmooth signals [39, 40], it is adopted to construct a wavelet filter bank in the predefined frequency band for filtering. The different frequency components of the vibration acceleration signal and the corresponding unimodal vibration response are extracted. The modal response of each order decomposed by the adopted modal decomposition procedure is visualized in Figure 8. It can be noticed that the VIV components of long suspenders are mainly concentrated on a few dominant frequencies.

The correlations between the frequency and the wind speed are shown in Figure 9. The color bar represents the maximum root-mean-square (RMS) acceleration of vibration modes. The modes with acceleration RMS below 3.0 m/s^2 are neglected. For the two sets of long suspenders, the main components of VIVs involved are around 12 Hz and 8 Hz , respectively. They correspond to the points with the maximum of accelerations RMS. Multimodes are fitted using linear functions, and error bands are labeled. Their R-

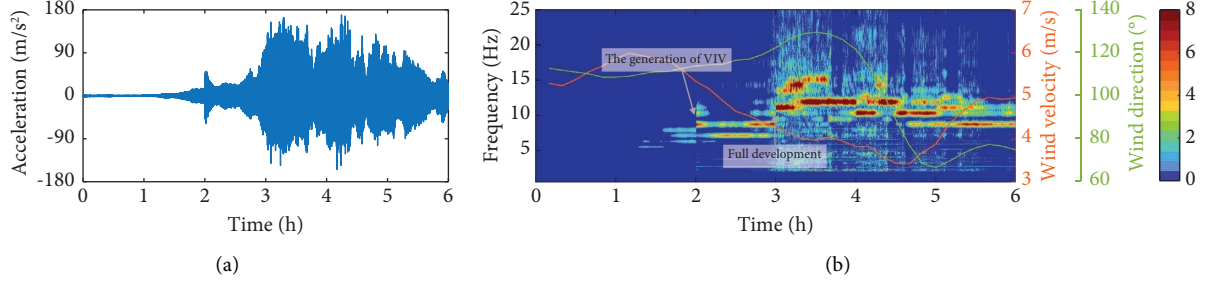


FIGURE 5: Typical acceleration time history and the corresponding time-frequency diagrams of S3. (a) Acceleration time history of S3. (b) Time-frequency diagrams by CWT of S3.

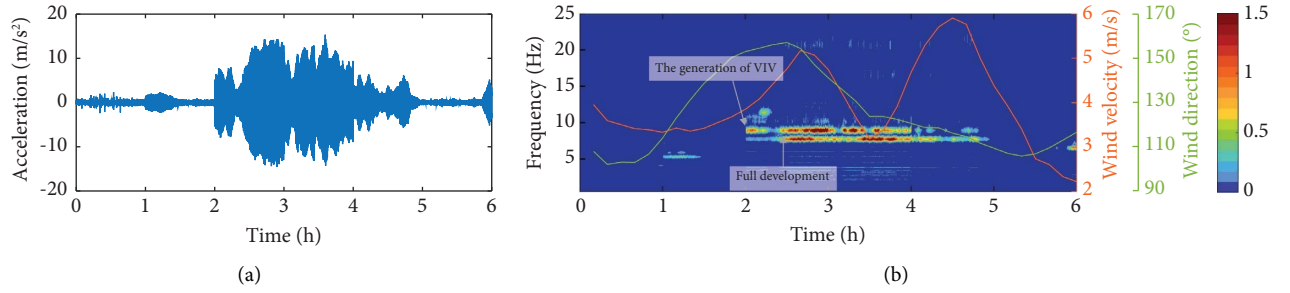


FIGURE 6: Typical acceleration time history and the corresponding time-frequency diagrams of S2. (a) Acceleration time history of S2. (b) Time-frequency diagrams by CWT of S2.

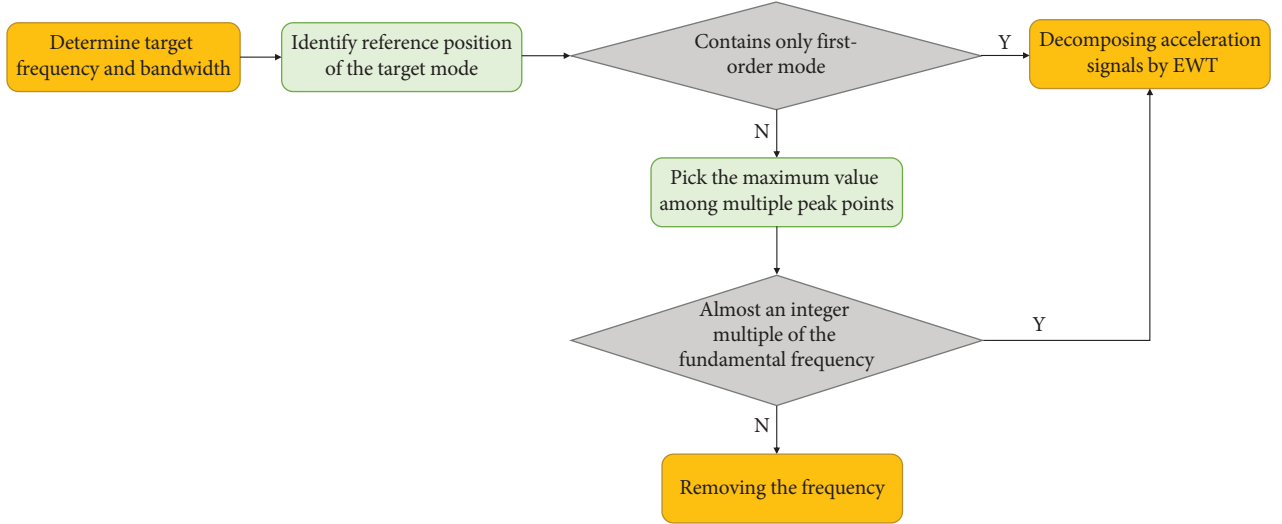


FIGURE 7: The flowchart of modal decomposition.

squared values were as high as 84.91% and 85.15%, respectively. These also confirm the theoretical linear relationship of frequency with mean wind speed.

For the phenomenon of vortex shedding, it can be described in terms of the nondimensional number, the Strouhal number:

$$St = \frac{fd}{v}, \quad (1)$$

where f is the frequency of full cycles of vortex shedding, d is the suspender diameter, and v is the wind velocity. In addition, the Strouhal number St depends on the object

geometry and the Reynolds number [41]. The Reynolds number in the assumed incompressible flow is

$$Re = \frac{\rho vd}{\mu}, \quad (2)$$

where μ is the dynamic viscosity and ρ is the density of the fluid. At an air density of 1.214 kg/m^3 and a viscosity of 1.8×10^{-5} , the Reynolds number distribution for wind speeds in the 1–15 m/s range is 1.1×10^4 – 3.8×10^4 . This result corresponds to the subcritical regime [42]. The maximum accelerations of the two long suspenders were found at $Re = 2.3 \times 10^4$ and $Re = 1.3 \times 10^4$, respectively.

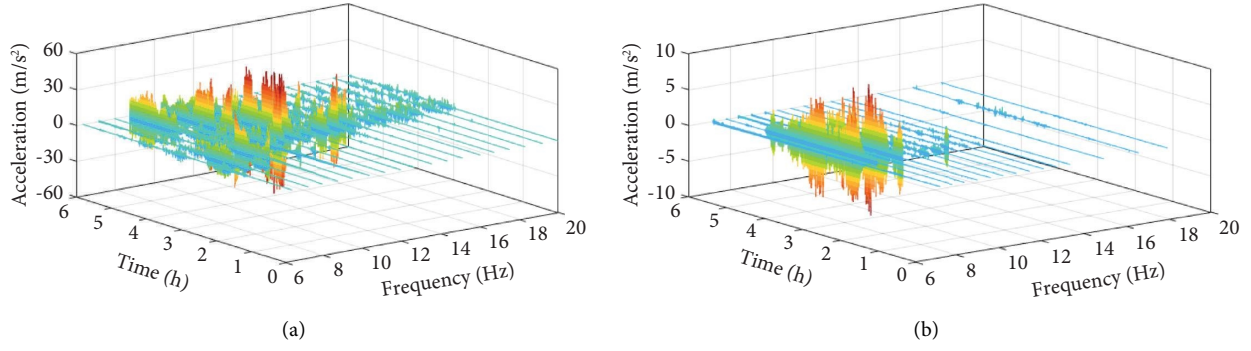


FIGURE 8: Signal decomposition of typical VIV samples. (a) S3. (b) S2.

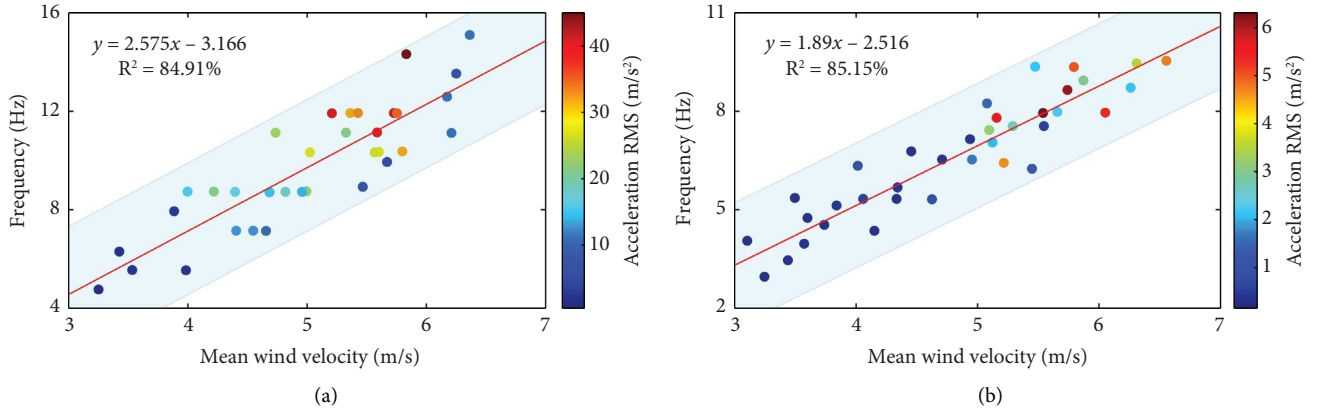


FIGURE 9: The correlation between the frequency and the wind speed. (a) S3. (b) S2.

Variations in the near wake of the long suspenders are suspected to be responsible for the development of the VIV based on the Reynolds number effect [41, 43].

It is often seen in related studies that the St of the cylinder is generally taken to be 0.2. However, in this study, a certain discrepancy was found between the frequency and the wind speed (Figure 9). This phenomenon is also evident in other studies [10, 37]. The relationship between St and Re for a cylinder was obtained after extensive experimentation [42–46]. In this study, it is difficult to determine the value of St frivolously with only one set of samples, and a large amount of data is needed to further determine the value of St subsequently.

3.2. Damping Estimation of Long Suspenders. To investigate the causes of high-frequency vortex-excited vibrations in long suspenders, modal damping ratios were estimated by on-site tests. Generally, it is not easy to extract accurate damping ratio from SHM. For closely spaced modes, the predicament becomes magnified. An automatic modal parameter identification approach was used to identify the damping ratio of the long suspenders [47]. The approach takes the modal validation criteria (MVC) as the eigenvector of each modal point and integrates principal component analysis, k-mean cluster, and hierarchical clustering algorithms to extract the effective modal parameters. The main concepts of the approach are as follows:

- (1) Calculate all the MVC values corresponding to each point in the stability diagram and form an eigenvector from the obtained MVC values.
- (2) Initial filtering of some spurious modes based on thresholds in preset frequency bands.
- (3) The high-dimensional eigenvectors of the remaining modal points were subjected to principal component analysis, and the first-order principal components were extracted for subsequent analyses.
- (4) The k-mean clustering method was used to classify the remaining modalities in the stability diagram into two groups using the Euclidean biparadigm for the distance metric. One group with larger central values was considered the spurious modes, and only the group with smaller central coordinate values was retained for subsequent analysis.
- (5) Based on the relationship between the number of truncated clusters of the hierarchical tree and the number of valid modes finally obtained, the optimal number of truncated clusters n of the hierarchical tree is determined.
- (6) Hierarchical clustering method was used to divide the remaining modal points into n groups and count the number of modal points in each group, whereby k-mean clustering ($k=2$) was used to divide the n groups of modal points into two categories. The one

containing fewer components was considered to be the spurious modal group, and the mean and variance of the remaining several groups of modal parameters were the final extracted.

The damping ratio estimates for acceleration RMS less than 0.3 m/s^2 were removed. It is evident from Figure 10 that the distribution of damping ratios for both sets of long suspenders has a large dispersion. The damping ratio of S3 in the frequency band from 7 Hz to 16 Hz is about 0.02% to 0.25%. S2 has a damping ratio of about 0.015% to 0.26% in the 6 Hz to 12 Hz. The frequency band is basically the area where high-frequency VIVs of the long suspenders occur. The correlation between the damping ratio and the acceleration RMS of the long suspenders is shown in Figure 11. As the vibration frequency increases, the modal damping ratio of the long suspenders decreases rapidly. The vibration acceleration is mainly concentrated at the dominant frequency. As the damping ratio decreases, the vibration amplitudes of the long suspenders rapidly increase. After that, the vibration amplitude tends to decay, which may be due to the fact that the VIVs of these order modes of the suspender are not excited in this environment.

3.3. Analysis of Vibration Response versus Wind Speed and Wind Direction. To further explore the effect of wind speed and wind direction on the VIVs of the long suspenders, the lock-in phenomenon was investigated. Figures 12 and 13 show the correlation between the acceleration RMS of the monitored long suspenders and the wind speed and wind direction. The vibrations of S3 and S2 occur mainly in the range of mean wind speeds of $2.0 \text{ m/s} \sim 7.0 \text{ m/s}$ and $3.0 \text{ m/s} \sim 6.0 \text{ m/s}$, respectively. The maximum acceleration RMS of the VIV in each mode is generally recorded in the range of $4.5 \text{ m/s} \sim 7.0 \text{ m/s}$ and $3.2 \text{ m/s} \sim 5 \text{ m/s}$ of mean wind speed. In addition, the VIVs of both sets of suspenders were mainly concentrated in the wind direction range from 90° to 120° .

The range of mean wind speeds corresponding to the maximum acceleration RMS decreases with increasing long suspender vibration frequency until no lock-in occurs. The vortex shedding frequency is constant for all wind speeds in the interval due to the effect of object deformation on the flow. When the flow affects the object motion, the object motion in turn affects the flow until it creates the locked-in [42, 43]. In other words, the vortex shedding frequency is simultaneous with the vibration frequency of the object. Since the relatively low modal damping, significant vortex characteristics and lock-in phenomena are observed on the long suspenders.

4. VIV Mitigation of Long Suspenders

4.1. The Effect of Damping Devices on Long Suspenders. To alleviate the detrimental effects of VIVs on the long suspenders and to provide additional damping, a novel frequency-adjustable Stockbridge damper was designed to control multimodal VIV, which is in essence a tuned mass damper (TMD). The damping mechanism and damping

effectiveness of the TMD have been widely reported [48–50]. Unlike existing conventional Stockbridge damper, they consist of four mass blocks attached to the long suspenders. The mounted mass blocks are of an asymmetrical design, which allows for a wider frequency range to be covered. The frequency-adjustable Stockbridge dampers are available prefabricated and fully assembled. The intrinsic frequency is regulated by adjusting the distance between the mass block and the anchorage. The dampers were welded to ensure durability and prevent loosening of the bolts. The design parameters as well as the optimization methodology are shown in Table 2 and Reference [51]. Figure 14 shows the Stockbridge damper installed on long suspenders above the deck on the NQYRB. To assess the performance of the dampers under strong wind conditions, measured data were collected from September 4, 2023, to September 22, 2023. The vibration characteristics of two sets of long suspenders were compared before and after the mounting of the dampers.

The representative example of the acceleration of the long suspenders is shown in Figure 15 for the condition where the Stockbridge dampers are installed. The occurrence frequency and acceleration amplitude of VIVs in the long suspenders were significantly reduced compared to the precontrol period. The acceleration amplitudes of the S3 and S2 were reduced from the previous 180 m/s^2 and 20 m/s^2 to below 8 m/s^2 and 4 m/s^2 , respectively, and the acceleration damping rate even reached 80%. Although the comparison is not time-consistent, it highlights to some extent the damping efficiency of the Stockbridge dampers.

4.2. The Effect of Damping Devices on Local Deck Vibration. For further analysis of the effect of the VIVs of long suspenders on the deck, the vibration characteristic of the deck during the VIVs was investigated using the measured data from the pickups installed on the decks. Figures 16 and 17 show the results of the time-frequency analysis of the deck vibrations corresponding to the S3 and S2. As shown in the figure, during the VIVs of the long suspenders, strong vibrations with an amplitude of more than 2 m/s^2 were generated on the deck.

From Figure 16, it can be seen that at the time of generation of VIVs of S3, the vibration of the girder tends to increase significantly. It then gradually decreases to the amplitude before the S3 vibration. In the time-frequency diagram, a clear high-frequency component of the long suspender VIVs can be observed, which is most likely due to the multimodal VIVs of the long suspenders. There are also some low-frequency components below 5 Hz, which may be due to wind and vehicle loads. In Figure 17, the relatively small vibration amplitude of S2 results in low high-frequency vibration energy in the deck. Instead, it is mainly concentrated below 5 Hz, and this component is basically caused by external sources (reciprocating heavy vehicles, wind loads, etc.). To further clarify the influence of long suspender VIVs on local deck vibration, the vibration transfer function was studied, as shown in Figure 18. It is clear that the VIV of the long suspenders has a direct

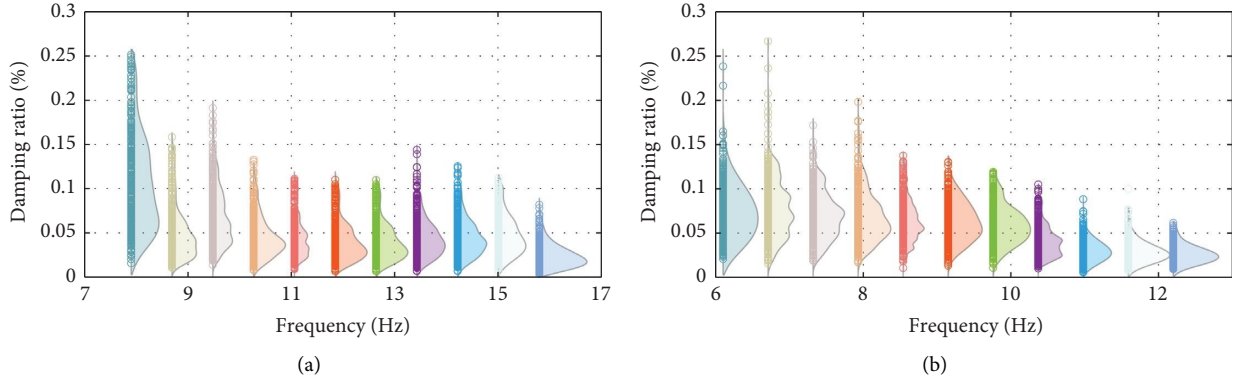


FIGURE 10: Evaluation of modal damping ratio of long suspenders. (a) S3. (b) S2.

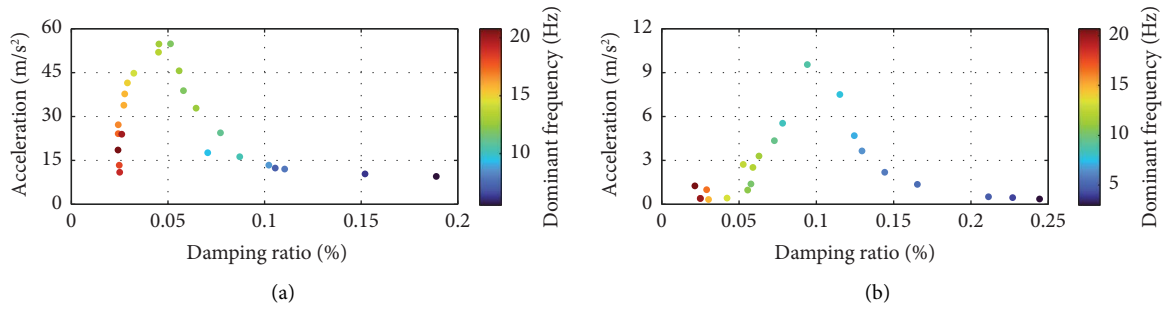


FIGURE 11: The correlation between the damping ratio and the acceleration RMS. (a) S3. (b) S2.

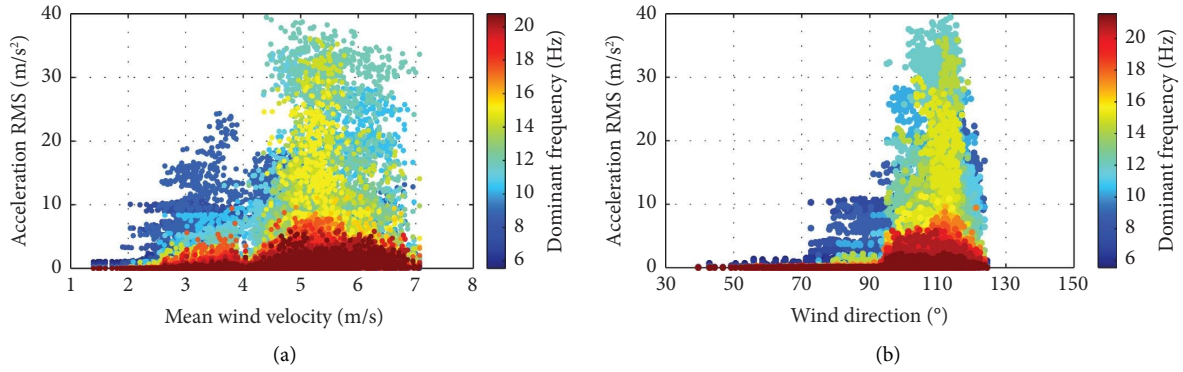


FIGURE 12: The acceleration RMS versus the mean wind velocity and wind direction of S3. (a) The acceleration RMS versus mean wind velocity. (b) The acceleration RMS versus wind direction.

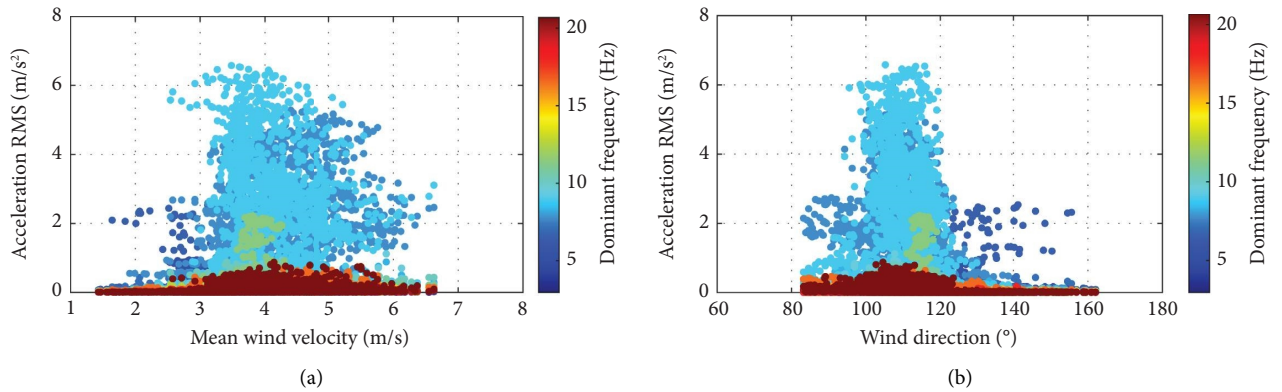


FIGURE 13: The acceleration RMS versus the mean wind velocity and wind direction of S2. (a) The acceleration RMS versus mean wind velocity. (b) The acceleration RMS versus wind direction.

TABLE 2: Design parameters for Stockbridge dampers installed in S2.

Number	S2-1			S2-2		
	Mass (kg)	Frequency (Hz)	Damping ratio (%)	Mass (kg)	Frequency (Hz)	Damping ratio (%)
1	9.9	8.95	2.08	9.9	9.03	1.86
2	9.4	10.21	2.17	9.4	10.34	2.23
3	8.8	10.95	1.98	8.8	11.04	1.76
4	7.6	11.63	2.01	7.6	11.55	2.07
5	6.7	12.47	2.12	6.7	12.35	2.26
6	6.7	13.40	1.81	6.7	13.54	2.26
7	6.7	13.94	1.27	6.7	14.15	1.68
8	5.9	16.61	1.06	5.9	16.81	1.02



FIGURE 14: The Stockbridge dampers installed in long suspenders.

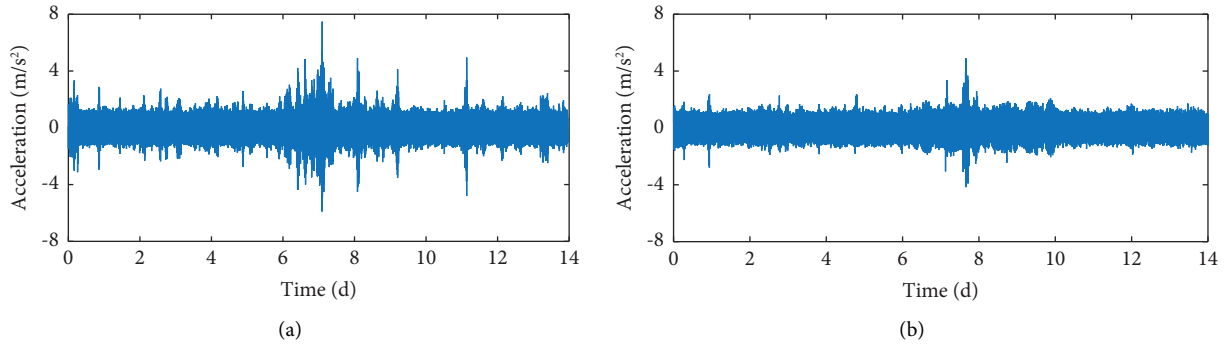


FIGURE 15: Typical acceleration time histories of long suspenders. (a) S3. (b) S2.

correlation with the vibration of the deck in the vicinity of their anchorage points. Of course, this observation needs to be further determined by vibration control of the long suspenders.

Figures 19 and 20 show the results of the typical acceleration time histories, and the PSD of the girder vibrations collected by the pickups corresponding to the S3 and S2

after damping. Compared to Figures 16 and 17, the amplitude of the vibration acceleration of the deck was significantly reduced from the previous 2 m/s^2 to 0.25 m/s^2 , retaining only roughly 12.5%. The PSD of the deck vibration is significantly reduced due to the VIV of the long suspenders being suppressed. The original high-frequency vibration component of the deck was significantly reduced

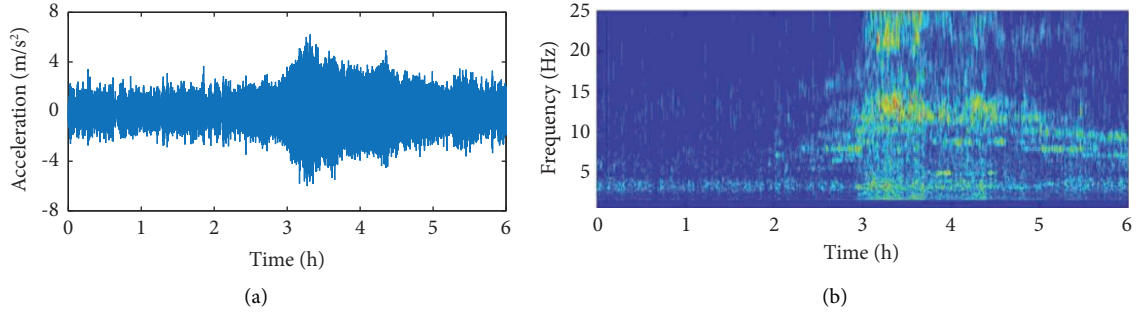


FIGURE 16: Typical acceleration time histories and the corresponding time-frequency diagrams of the deck near S3. (a) Acceleration time history. (b) Time-frequency diagrams of the deck.

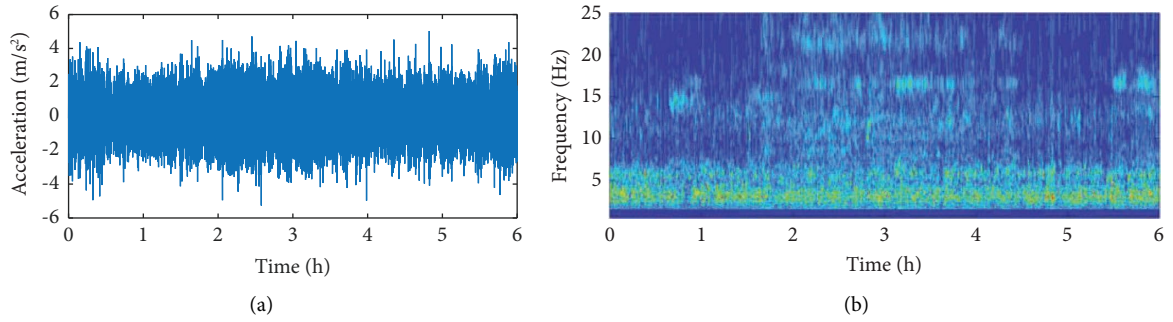


FIGURE 17: Typical acceleration time histories and the corresponding time-frequency diagrams of the deck near S2. (a) Acceleration time history. (b) Time-frequency diagrams of the deck.

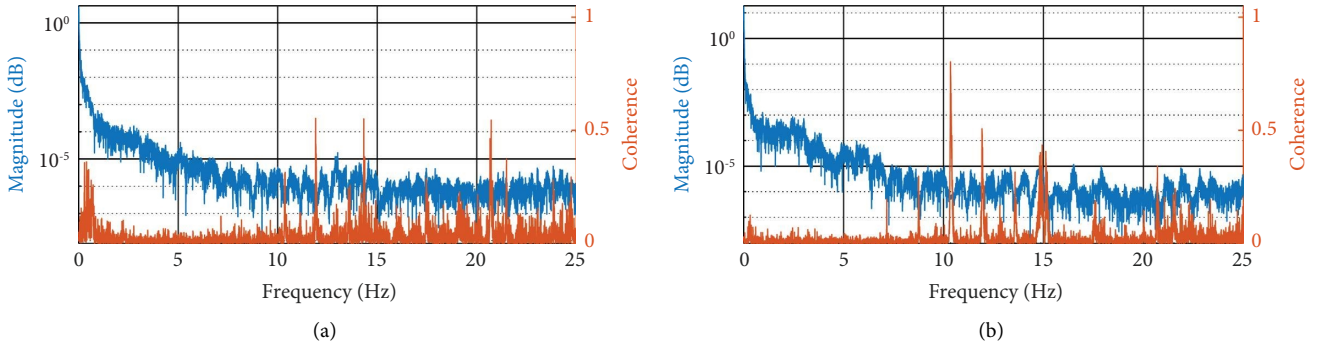


FIGURE 18: Frequency response function of long suspender VIV to local deck vibration. (a) S3. (b) S2.

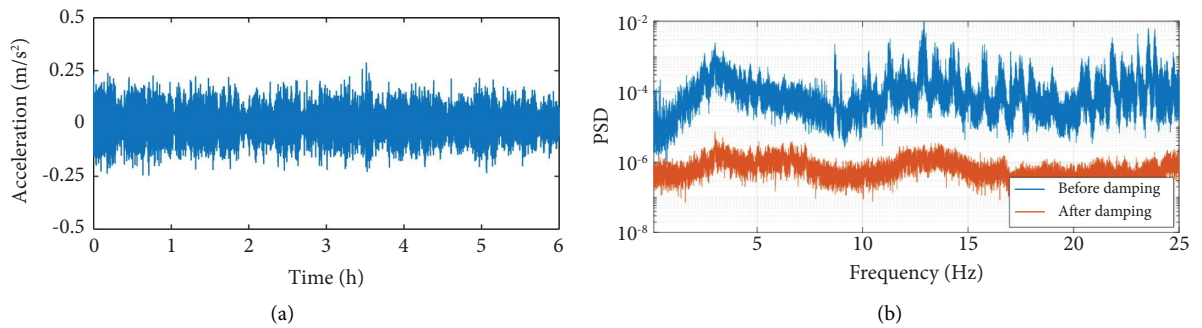


FIGURE 19: Typical acceleration time histories and the PSD of the deck near S3. (a) Acceleration time history. (b) PSD of the deck vibration.

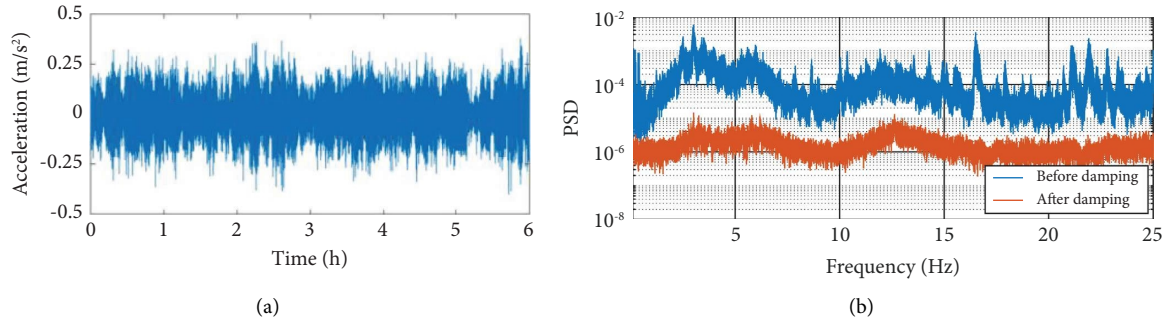


FIGURE 20: Typical acceleration time histories and the PSD of the deck near S2. (a) Acceleration time history. (b) PSD of the deck.

with the installation of the damping device for the long suspenders. This further confirms the earlier conclusion that long suspender vibrations are the main cause of local deck vibration. As for the remaining small amount of vibration, it can be determined to be caused by external sources.

5. Conclusions

This study documented the VIV events of different long suspenders of a long-span suspension bridge. The focus is on the vibrational characteristics and excitation mechanisms, as well as the effect of long suspender VIVs on the local deck acceleration based on the field monitoring data. The main findings of this study are summarized as follows:

- (1) The VIVs with frequencies ranging from 8 Hz to 20 Hz were detected continuously over a wide range of wind speeds in both sets of long suspenders. The most dangerous vibrations occurred at 15 Hz and 10 Hz, respectively.
- (2) For two sets of long suspenders with different tension and sizes, the S3 has more vibration modes. During VIVs, it has a greater vibration acceleration and a more pronounced effect on the deck. Due to the relatively low modal damping, significant vortex characteristics and lock-in phenomena are foreseen on the long suspenders.
- (3) Through the time-frequency analysis of the deck before and after damping, it was clarified that the vibration of the deck was mainly caused by the multimodal VIVs of the long suspenders, rather than the vehicle and wind load. The results may offer insights into estimating the performance of long suspender dampers of the whole bridge using the deck vibration data monitored by the SHM system, and for identifying and locating the anomalies.
- (4) Since the installation of the Stockbridge dampers on the long suspenders, the excessive frequency of the VIV has been perfectly controlled. The vibration amplitude has been significantly reduced, with almost no dangerous vibrations occurring. The costs required for suspender maintenance are significantly reduced.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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References

- [1] J. Zhang, J. Prader, K. A. Grimmelmsan, F. Moon, A. E. Aktan, and A. Shama, "Experimental vibration analysis for structural identification of a long-span suspension bridge," *Journal of Engineering Mechanics*, vol. 139, no. 6, pp. 748–759, 2013.
- [2] O. Øiseth, A. Rönquist, and R. Sigbjörnsson, "Simplified prediction of wind-induced response and stability limit of slender long-span suspension bridges, based on modified quasi-steady theory: a case study," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 98, no. 12, pp. 730–741, 2010.
- [3] E. Laursen, N. Bitsch, and J. E. Andersen, "Analysis and mitigation of large amplitude cable vibrations at the Great Belt East Bridge," *IABSE Conference, Copenhagen 2006: Operation, Maintenance and Rehabilitation of Large Infrastructure Projects, Bridges and Tunnels*, vol. 91, no. 3, pp. 64–71, 2006.
- [4] S. Kashima, Y. Yanaka, S. Suzuki, and K. Mori, "Monitoring the Akashi Kaikyo bridge: first experiences," *Structural Engineering International*, vol. 11, no. 2, pp. 120–123, 2001.
- [5] Y. Deng, S. Li, M. Zhang, X. Lei, and Z. Chen, "Wake-induced vibrations of the hangers of the Xihoumen bridge," *Journal of Bridge Engineering*, vol. 26, no. 10, Article ID 05021012, 2021.
- [6] X. W. Ye, Y. Ding, and H. P. Wan, "Probabilistic forecast of wind speed based on Bayesian emulator using monitoring

- data," *Structural Control and Health Monitoring*, vol. 28, no. 1, Article ID e2650, 2021.
- [7] X. G. Hua, Z. Q. Chen, X. Lei, Q. Wen, and H. W. Niu, "Monitoring and control of wind-induced vibrations of hanger ropes of a suspension bridge," *Smart Structures and Systems, an International Journal*, vol. 23, no. 6, pp. 683–693, 2019.
 - [8] H. Gjelstrup, C. T. Georgakis, and A. Larsen, "An evaluation of iced bridge hanger vibrations through wind tunnel testing and quasi-steady theory," *Wind and Structures An International Journal*, vol. 15, no. 5, pp. 385–407, 2012.
 - [9] W. Chen, D. Gao, H. Li, and H. Hu, "Wake-flow-induced vibrations of vertical hangers behind the tower of a long-span suspension bridge," *Engineering Structures*, vol. 169, pp. 188–200, 2018.
 - [10] D. Cantero, O. Øiseth, and A. Rønquist, "Indirect monitoring of vortex-induced vibration of suspension bridge hangers," *Structural Health Monitoring*, vol. 17, no. 4, pp. 837–849, 2018.
 - [11] S. O. Hansen, "Vortex-induced vibrations—the Scruton number revisited," *Proceedings of the Institution of Civil Engineers-Structures and Buildings*, vol. 166, no. 10, pp. 560–571, 2013.
 - [12] T. Wu and A. Kareem, "An overview of vortex-induced vibration (VIV) of bridge decks," *Frontiers of Structural and Civil Engineering*, vol. 6, no. 4, pp. 335–347, 2012.
 - [13] X. Wu, F. Ge, and Y. Hong, "A review of recent studies on vortex-induced vibrations of long slender cylinders," *Journal of Fluids and Structures*, vol. 28, pp. 292–308, 2012.
 - [14] J. A. Main and N. P. Jones, "Evaluation of viscous dampers for stay-cable vibration mitigation," *Journal of Bridge Engineering*, vol. 6, no. 6, pp. 385–397, 2001.
 - [15] E. de Sá Caetano, "Cable vibrations in cable-stayed bridges," *IABSE*, vol. 9, 2007.
 - [16] C. H. Williamson and R. Govardhan, "Vortex-induced vibrations," *Annual Review of Fluid Mechanics*, vol. 36, no. 1, pp. 413–455, 2004.
 - [17] W. L. Chen, Q. Q. Zhang, H. Li, and H. Hu, "An experimental investigation on vortex induced vibration of a flexible inclined cable under a shear flow," *Journal of Fluids and Structures*, vol. 54, pp. 297–311, 2015.
 - [18] Z. Liu, J. Shen, S. Li, Z. Chen, Q. Ou, and D. Xin, "Experimental study on high-mode vortex-induced vibration of stay cable and its aerodynamic countermeasures," *Journal of Fluids and Structures*, vol. 100, Article ID 103195, 2021.
 - [19] F. Di, L. Sun, and L. Chen, "Suppression of vortex-induced high-mode vibrations of a cable-damper system by an additional damper," *Engineering Structures*, vol. 242, Article ID 112495, 2021.
 - [20] F. J. Huera-Huarte and P. W. Bearman, "Wake structures and vortex-induced vibrations of a long flexible cylinder—part 1: dynamic response," *Journal of Fluids and Structures*, vol. 25, no. 6, pp. 969–990, 2009.
 - [21] G. Diana, F. Resta, M. Belloli, and S. Muggiasca, "Experimental analysis on vortex induced vibration of a long flexible cylinder," in *Proceedings of 6th International Symposium on Cable Dynamics*, pp. 1–8, September 2005.
 - [22] F. Xu, Z. Ma, H. Zeng et al., "A new method for studying wind engineering of bridges: large-scale aeroelastic model test in natural wind," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 202, Article ID 104234, 2020.
 - [23] X. G. Hua, Z. Q. Chen, W. Chen, H. W. Niu, and Z. W. Huang, "Investigation on the effect of vibration frequency on vortex-induced vibrations by section model tests," *Wind and Structures*, vol. 20, no. 2, pp. 349–361, 2015.
 - [24] Z. Kang, C. Zhang, R. Chang, and G. Ma, "A numerical investigation of the effects of Reynolds number on vortex-induced vibration of the cylinders with different mass ratios and frequency ratios," *International Journal of Naval Architecture and Ocean Engineering*, vol. 11, no. 2, pp. 835–850, 2019.
 - [25] J. Lim, S. Kim, and H. K. Kim, "Using supervised learning techniques to automatically classify vortex-induced vibration in long-span bridges," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 221, Article ID 104904, 2022.
 - [26] J. Park, S. Kim, and H. K. Kim, "Effect of gap distance on vortex-induced vibration in two parallel cable-stayed bridges," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 162, pp. 35–44, 2017.
 - [27] Y. Bao, Z. Chen, S. Wei, Y. Xu, Z. Tang, and H. Li, "The state of the art of data science and engineering in structural health monitoring," *Engineering*, vol. 5, no. 2, pp. 234–242, 2019.
 - [28] J. Ou and H. Li, "Structural health monitoring in mainland China: review and future trends," *Structural Health Monitoring*, vol. 9, no. 3, pp. 219–231, 2010.
 - [29] L. Li, H. Zhou, H. Liu, C. Zhang, and J. Liu, "A hybrid method coupling empirical mode decomposition and a long short-term memory network to predict missing measured signal data of SHM systems," *Structural Health Monitoring*, vol. 20, no. 4, pp. 1778–1793, 2021.
 - [30] C. Xing, H. Wang, A. Li, and Y. Xu, "Study on wind-induced vibration control of a long-span cable-stayed bridge using TMD-type counterweight," *Journal of Bridge Engineering*, vol. 19, no. 1, pp. 141–148, 2014.
 - [31] C. Xing, H. Wang, A. Li, and Y. Xu, "Study on wind-induced vibration control of a long-span cable-stayed bridge using TMD-type counterweight," *Journal of Bridge Engineering*, vol. 19, no. 1, pp. 141–148, 2014.
 - [32] A. Larsen, K. G. Andersen, and A. Jamal, "Wind induced hanger vibrations—the Hålogaland suspension bridge," *Structural Engineering International*, vol. 32, no. 1, pp. 62–70, 2022.
 - [33] Y. Duan, Y. Q. Ni, H. Zhang, B. F. Jr, J. M. Ko, and Y. Fang, "Design formulas for vibration control of taut cables using passive MR dampers," *Smart Structures and Systems, An International Journal*, vol. 23, no. 6, pp. 521–536, 2019.
 - [34] Z. Q. Chen, X. Y. Wang, J. M. Ko et al., "MR damping system for mitigating wind-rain induced vibration on Dongting Lake Cable-Stayed Bridge," *Wind and Structures*, vol. 7, no. 5, pp. 293–304, 2004.
 - [35] S. W. Or, Y. F. Duan, Y. Q. Ni, Z. H. Chen, and K. H. Lam, "Development of magnetorheological dampers with embedded piezoelectric force sensors for structural vibration control," *Journal of Intelligent Material Systems and Structures*, vol. 19, no. 11, pp. 1327–1338, 2008.
 - [36] W. Wang, X. Hua, X. Wang, Z. Chen, and G. Song, "Numerical modeling and experimental study on a novel pounding tuned mass damper," *Journal of Vibration and Control*, vol. 24, no. 17, pp. 4023–4036, 2018.
 - [37] S. Kim, S. Kim, and H. K. Kim, "High-mode vortex-induced vibration of stay cables: monitoring, cause investigation, and mitigation," *Journal of Sound and Vibration*, vol. 524, Article ID 116758, 2022.
 - [38] H. Zhang, J. Mao, H. Wang et al., "A novel acceleration-based approach for monitoring the long-term displacement of bridge cables," *International Journal of Structural Stability and Dynamics*, vol. 23, no. 05, Article ID 2350053, 2023.

- [39] D. Li, J. H. Nie, H. Wang, J. B. Yan, C. X. Hu, and P. Shen, "Damage location, quantification and characterization of steel-concrete composite beams using acoustic emission," *Engineering Structures*, vol. 283, Article ID 115866, 2023.
- [40] M. Kedadouché, M. Thomas, and A. J. M. S. Tahan, "A comparative study between empirical wavelet transforms and empirical mode decomposition methods: application to bearing defect diagnosis," *Mechanical Systems and Signal Processing*, vol. 81, pp. 88–107, 2016.
- [41] C. Norberg, "Fluctuating lift on a circular cylinder: review and new measurements," *Journal of Fluids and Structures*, vol. 17, no. 1, pp. 57–96, 2003.
- [42] E. Simiu and R. H. Scanlan, *Wind Effects on Structures: Fundamentals and Applications to Design*, John Wiley, New York, NY, USA, 1996.
- [43] K. Raghavan and M. M. Bernitsas, "Experimental investigation of Reynolds number effect on vortex induced vibration of rigid circular cylinder on elastic supports," *Ocean Engineering*, vol. 38, no. 5-6, pp. 719–731, 2011.
- [44] C. Norberg, *Reynolds Number and Freestream Turbulence Effects on the Flow and Fluid Forces for a Circular cylinder in Cross Flow*, 1987.
- [45] O. Posdziech and R. Grundmann, "A systematic approach to the numerical calculation of fundamental quantities of the two-dimensional flow over a circular cylinder," *Journal of Fluids and Structures*, vol. 23, no. 3, pp. 479–499, 2007.
- [46] S. J. Baek and H. J. Sung, "Numerical simulation of the flow behind a rotary oscillating circular cylinder," *Physics of Fluids*, vol. 10, no. 4, pp. 869–876, 1998.
- [47] J. X. Mao, H. Wang, Y. G. Fu, and B. F. Spencer Jr, "Automated modal identification using principal component and cluster analysis: application to a long-span cable-stayed bridge," *Structural Control and Health Monitoring*, vol. 26, no. 10, Article ID e2430, 2019.
- [48] L. Zuo and S. A. Nayfeh, "Optimization of the individual stiffness and damping parameters in multiple-tuned-mass-damper systems," *Journal of Vibration and Acoustics*, vol. 127, no. 1, pp. 77–83, 2005.
- [49] S. Nagarajaiah and E. Sonmez, "Structures with semiactive variable stiffness single/multiple tuned mass dampers," *Journal of Structural Engineering*, vol. 133, no. 1, pp. 67–77, 2007.
- [50] Z. Wang, Y. Liu, H. Gao, Z. Cheng, H. Wang, and Y. Xu, "Multi-mode vibration control of super-long stay cables with negative stiffness and Stockbridge dampers," *International Journal of Structural Stability and Dynamics*, Article ID 2550001, 2024.
- [51] H. Gao, H. Wang, J. Mao, X. Guo, X. Su, and M. Lepidi, "Field monitoring and control for vortex-induced vibration of hanger cables in suspension bridge," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 248, Article ID 105712, 2024.